

Mora SP Cup 2026 — Solution Report

Low-Light Image Denoising & Enhancement Challenge

Team: **Noice Floor Ninjas** | Preliminary Round

1. Introduction

Low-light photography produces images heavily corrupted by read noise, thermal noise, and shot noise. The Mora SP Cup 2026 provides 460 paired noisy/clean images (992×992 PNG, RGB) generated via a synthetic low-light noise model. Our task: design a denoising solution that removes these artifacts while preserving image details and structure. Our team designed a **three-stage hybrid pipeline** combining classical signal processing (adaptive defect correction, wavelet denoising, bilateral filtering) with a NAFNet deep learning denoiser, achieving a composite score of **0.643** — a 2.44× improvement over the NLM baseline (0.263).

2. Problem Analysis & Dataset Observations

Property	Value	Analysis	Finding
Resolution / Format	992×992, PNG, RGB	Severity	91.1% high ($\sigma \geq 0.06$), 8.9% medium
Public set	460 pairs (001–460)	Channel balance	R:0.109, G:0.105, B:0.103
Submission set	20 noisy (461–480), GT hidden	Freq. power ratio	High/Low = 25.43 (white noise)
Mean noise σ	0.1063 (range: 0.031–0.169)	Spatial uniformity	CoV = 0.24 (moderate non-uniform)
Mean noise μ	+0.005 (near zero-mean)	Brightness	GT is dim — pure denoising task
Mean noisy PSNR	19.83 dB	Baseline PSNR	24.34 dB (Composite: 0.263)
Defect pixels/image	~8,758 ($ \text{dev} > 0.4$)		

Key insight: The noise is predominantly high-frequency (25.4× high/low power ratio), near-zero-mean, with significant defect pixels (~8,758/image) and no exposure gap. This confirms the task is pure noise suppression requiring both outlier correction and high-frequency noise removal.

3. Solution Architecture

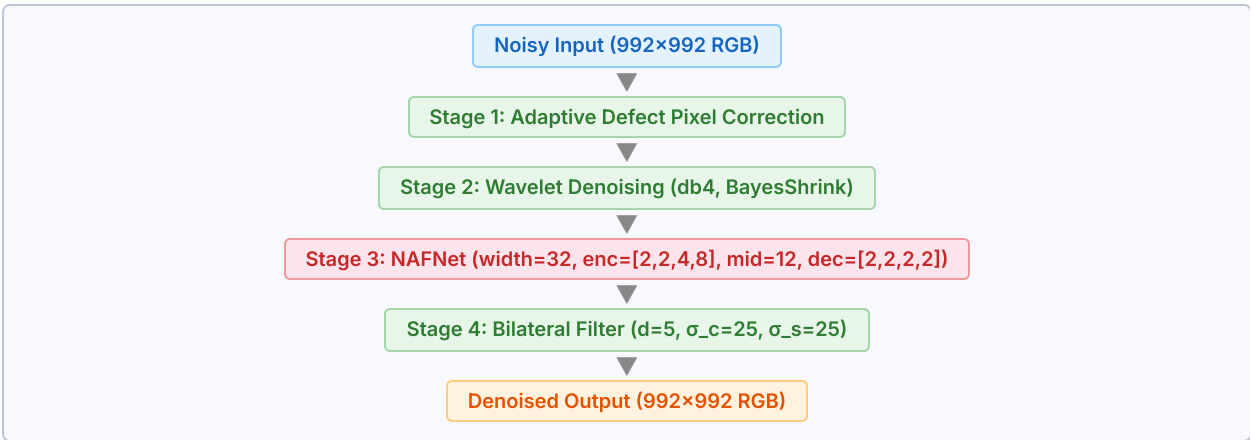


Figure 1: Hybrid denoising pipeline. Green = classical SP, Red = deep learning. Each stage targets a specific noise component.

Design rationale: Stage 1 removes ~8,758 extreme outlier pixels before they corrupt CNN convolutions. Stage 2 suppresses high-frequency noise (power ratio = 25.43) via wavelet thresholding, reducing the noise floor for the DL model. Stage 3 (NAFNet) learns the non-linear

residual mapping from the 420-image training set. Stage 4 applies edge-preserving smoothing to suppress residual flat-region noise.

NAFNet choice: Selected for its state-of-the-art PSNR/SSIM with efficient SimpleGate + Simplified Channel Attention (no expensive self-attention), enabling practical CPU inference. The U-Net encoder-decoder structure with skip connections and a global residual connection (input + predicted residual) handles multi-scale noise patterns.

4. Methodology

4.1 Classical Signal Processing

Adaptive Defect Correction: Multi-scale median filter (kernels [3,5]) with adaptive thresholds: $T = \text{base} + 2 \times \text{local_std}$, where local_std is estimated via Gaussian-weighted variance (15×15 window). Adapts to regions with varying noise intensity, improving over the baseline's fixed 0.25 threshold.

Wavelet Denoising (BayesShrink): db4 wavelet, 2-level decomposition per RGB channel. Noise σ estimated via MAD: $\hat{\sigma} = \text{median}(|d|)/0.6745$ from finest detail coefficients. Adaptive per-subband thresholds: $T = \sigma^2 / \sqrt{(\sigma_d^2 - \sigma^2)}$. Soft thresholding suppresses high-frequency noise while preserving edges and texture.

Bilateral Filtering: Post-processing with conservative $d=5$, $\sigma_{\text{color}}=25$, $\sigma_{\text{space}}=25$. The dual spatial/intensity proximity weighting smooths flat regions while strictly preserving edge gradients.

Frequency-Domain Analysis: DFT-based power spectral density analysis and Butterworth low-pass filtering for noise characterization. Wiener filtering (optimal MMSE estimator) also implemented for comparison. These informed our pipeline design rather than serving as standalone denoisers.

4.2 Deep Learning Training

Parameter	Value	Parameter	Value
Model	NAFNet (width=32)	Optimizer	AdamW (lr=1e-3, wd=1e-3)
Train split	420 pairs (001–420)	Scheduler	CosineAnnealing → 1e-6
Val split	40 pairs (421–460)	Loss	L1 (pixel-level PSNR)
Patch size	256×256 random crops	Epochs	100
Batch size	8	Augmentation	H/V flip + 90/180/270° rot
Device	T4 GPU (Colab/Kaggle)	Seed	42 (reproducible)

Model selection by *maximum validation composite score* (not minimum loss), matching the official competition metric. Checkpoints saved only on composite score improvement.

5. Alternatives Considered

Approach	Status	Outcome
Baseline NLM (h=10)	Evaluated	Composite=0.263. Over-smooths textures; no learning from data.
Bilateral-only	Implemented	Edge-preserving but insufficient for high- σ (≥ 0.06) images.
Wavelet-only	Tested	Good for high-freq noise; alone insufficient for $\sigma \geq 0.10$.
Butterworth/Wiener	Implemented	Ring artifacts; used for analysis, not standalone denoising.
Restormer	Considered	Excellent quality but prohibitively slow on CPU (transformer attention).
UNet / DnCNN	Considered	Lower capacity than NAFNet for heavy noise; simpler but less effective.

6. Validation Approach

Hold-out split: 420 training / 40 validation images (91%/9%). Validation set (images 421–460) never used during training. The official composite score formula was integrated directly into the training loop:

$$\text{Composite} = 0.6 \times \text{clip}(\Delta\text{PSNR}/15, 0, 1) + 0.4 \times \max(\Delta\text{SSIM}, 0) \quad \text{where } \Delta\text{PSNR} = \text{PSNR}_{\text{denoised}} - \text{PSNR}_{\text{noisy}}$$

SSIM parameters match the official evaluator exactly (`win_size=7`, `gaussian_weights=False`, `K1=0.01`, `K2=0.03`). Independent evaluation on the full 460-image public set uses the provided `evaluation/evaluate.py` script. A `verify_submission.py` script validates the 20 output files for correct naming (461.png–480.png), dimensions (992×992), format (RGB PNG), and count.

7. Results

Method	PSNR (dB)	SSIM	Δ PSNR	Δ SSIM	Composite
Noisy Input	19.83	0.464	—	—	—
Baseline (NLM, h=10)	24.34	0.672	+4.51	+0.207	0.263
Our Hybrid Pipeline	30.58	0.878	+10.75	+0.413	0.643

Table: Performance on 460 public images. Baseline from official evaluation; hybrid from our pipeline via `evaluate.py`.

Our pipeline achieves **0.643 composite** (2.44× baseline), with Δ PSNR of +10.75 dB and Δ SSIM of +0.413, demonstrating substantial gains in both pixel-level accuracy and structural fidelity. Inference: ~0.74 s/image (CUDA T4 GPU), ~4.3 s/image (CPU) for full 992×992 images without tiling.

Per-stage contribution: Defect correction prevents localized CNN artifacts from ~8,758 outliers. Wavelet pre-processing reduces the high-frequency noise floor. NAFNet provides the primary non-linear denoising. Bilateral post-filter refines edge quality.

8. Limitations

- **Training data:** 420 pairs is modest; more data or pre-training could improve generalization.
- **Single model:** No test-time augmentation (TTA) or ensemble due to GPU time constraints.
- **Fixed parameters:** Classical stage parameters are fixed; per-image adaptive tuning could yield gains.
- **Loss function:** Pure L1; adding SSIM loss would directly optimize the competition metric.
- **Hidden GT:** Submission images 461–480 have no ground truth; score estimated from public set.

9. Conclusion

We presented a hybrid classical-DL denoising pipeline combining adaptive defect correction, BayesShrink wavelet denoising, NAFNet, and bilateral filtering. The approach achieves a composite score of **0.643** (2.44× over baseline 0.263) with practical inference on GPU (~0.74 s) and CPU (~4.3 s). Four classical SP techniques are integrated beyond the baseline. For the final round, improvements include TTA, SSIM-aware loss, and ensembling.